

QM 2023 Capstone Project – Milestone 4

Individual Contribution Statement

Name: Katrina Baiza

Team: 2nd Row Team

Date: May 1, 2026

1. Personal Contribution to Capstone Milestones

Milestone 1: Data Pipeline (Week 5)

For this milestone, I worked on cleaning and organizing the raw datasets so they were ready for analysis. I also helped fill out the AI AUDIT file, where we documented our AI usage.

Time spent: 5 hours

- **Key deliverables:** Clean, well-structured datasets and an updated AI AUDIT with clear use and prompts used.

Milestone 2: EDA Dashboard (Week 10)

I helped interpret the graphs made by other teammates and gather information. I also helped to write some of the economic summary.

Time spent: 5 hours

- **Key deliverables:** Data visualization interpretation and a written summary of findings and hypotheses.

Milestone 3: Econometric Models (Week 14)

I helped specify and interpret the differential effect on sentiment vs controls regression model. I also participated in robustness checks, such as removing COVID years and comparing results across firms of different sizes.

- **Time spent:** 5 hours
- **Key deliverables:** Interpretation report for Milestone 3 and supporting robustness analysis.

Milestone 4: Final Investment Memo (Week 14)

I contributed to drafting the executive summary and helped edit the final memo to improve clarity and professionalism. I also supported the team in preparing for the final presentation by editing the PowerPoint/final report.

- **Time spent:** 5 hours

- **Key deliverables:** Final investment memo and polished presentation slides.

Total Contribution:

- **Total hours:** 20 hours
- **Estimated workload share:** 30%
- **Role:** Communication lead and primary recorder of findings for both written work and the presentation.

2. One Defended Methodological Decision

I supported using a Difference-in-Differences (DiD) approach to examine how sentiment responds to major economic shocks relative to market and factor variables.

The goal of this method was to compare sentiment indicators (treated group) to market returns and risk factors (control group) before and after key events like the 2008 financial crisis and COVID-19. This allowed us to isolate whether sentiment behaved differently from other financial variables during periods of economic stress.

Our results showed that after COVID, the interaction between sentiment and the post COVID period was negative and statistically significant. This suggests that sentiment weakened more than market returns and factor controls during that time. In contrast, the post-2008 interaction was not statistically significant, indicating no clear differential effect during that period.

To validate this approach, we conducted robustness checks, including excluding the COVID crash window and running placebo tests before the 2008 crisis. The results remained consistent, and the placebo tests showed no strong effects, which supports the credibility of our identification strategy.

Our exploratory analysis and robustness checks showed that the relationship between sentiment and returns stayed positive across different lag structures (for example, coefficients of 0.0277 vs. 0.0078). This suggests the results are not dependent on a single lag choice. Even after excluding COVID years, the relationship remained consistent (coefficient = 0.0118), which further supports the reliability of our model.

3. One Key Limitation of Our Analysis

One limitation of our model is that it assumes certain REIT characteristics, such as property type exposure and leverage, do not change over time. In reality, these factors can shift, especially during major events like COVID.

If these changes are related to both returns and macroeconomic variables, they could bias our results. To address this in future research, we could include time-varying controls, use a dynamic panel model, or add interaction effects to better capture industry-wide changes.

4. AI Audit Notes

All AI tools used during the project were properly documented in the AI Audit Appendix for each milestone.

5. Self-Reflection

What I did well:

I was strong at interpreting our data and analysis into clear, meaningful insights about how sentiment affects the stock market. I also helped keep our research focused by consistently linking our findings back to our main research question.

What I could improve:

I could have spent more time going further into the research to better understand its broader implications. This would have allowed me to explore additional data sources and conduct more advanced robustness checks.

What I learned:

This project showed me how important data cleaning and alignment are, especially when working with multiple sources like financial data and sentiment surveys. I also learned how to evaluate whether sentiment acts as a predictor or a contrarian indicator using both data and economic reasoning.

6. Attestation

By submitting this statement, I confirm that all contributions listed are accurate and honestly reflect my work. I have not overstated my role or understated my teammates' contributions. I understand this may be used to adjust my individual grade, and I take full responsibility for my work.

Signature: Katrina Baiza

Date: May 1, 2026